

Exam in "Economics & Law"

Winter Semester 2021/2022

Please read carefully the following instructions:

- This is a written exam based on 4 questions.
 - Each answer awards a different amount of points, as stated in brackets at the end of the question. The total points sum up to 100.
- This is an open-book exam. You may bring with you any written/printed material but you are not allowed to use a computer.

Questions:

Q1. "GreenTech" is a firm that produces solar panels. GreenTech has been enjoying high profits for the past year and therefore decided to donate money to the election campaign of the minister of finance. However, two changes have occurred in the market: First, a breakthrough in the production of silicon, a key material used in the production of solar panels, has made it cheaper and more readily available. Second, a new government policy has been implemented, which required the installation of solar panels in every governmental office.

- (a) How should these events affect the equilibrium price and equilibrium quantity? **[10 points]**
- (b) Which economic mechanisms could explain the governmental policy mentioned in the question? **[10 points]**

Q2. A young lawyer and an old lawyer are sitting on a bench. A potential client passes by. If both lawyers run toward the client at the same time, they slip and fall down, so each of the lawyers gets a negative utility of -10. If only one lawyer approaches the client then he signs the client (whereas the other remains on the bench and gets zero). The old lawyer values signing the client at 20. The young lawyer values signing the client at 10. If no one approaches the client, both lawyers get a utility of zero.

- (a) What is the Nash Equilibrium (or Nash Equilibria) of this game? **[10 points]**
- (b) Is the equilibrium (or equilibria) Pareto Efficient? Is it Kaldor-Hicks efficient? **[5 points]**
- (c) Suppose that the young lawyer is faster than the old lawyer, and can therefore act first. What is then the equilibrium? **[5 points]**
- (d) Under the same assumption as in sub-question (c), suppose the old lawyer tells the young lawyer: "don't you dare approach the client! If you do, I will do so as well and we both slip and fall". Should the young lawyer worry about this threat? Explain your answer. **[5 points]**
- (e) If the two lawyers could bargain at zero transaction costs, what would be the outcome of the game? **[5 points]**

Q3. Jimmy is a judge in a criminal case. He decides to ask an AI-based chatbot what he recommends to do. The chatbot's output is the following:

"The defendant has long fingernails. It is well-known, that: 90% of people with long fingernails commit crimes; 10% of people with short fingernails do not commit crimes; and 0.1% of people have long fingernails. Furthermore, please note that 65% of judges using our services convict in these cases, will you do the same? By the way, there are two news articles about people with long fingernails who committed a crime (both of them robbed a bank). Also, you are our 25th user today. Thanks for using this Chatbot!"

Jimmy decides to convict the defendant and sentences him to 24 years in prison. The defendant wants to appeal and turns to you. Which arguments from behavioral economics could you use in your appeal (note: there is no need to use numerical calculations in your answer)? **[25 points]**

Q4. In the developing country of “DevState”, there are large demonstrations protesting the inequality between employers and employees. The president of DevState turns to his advisors for advice:

Advisor 1: We should improve the working conditions for our workers by limiting their working week to only four days.

Advisor 2: I agree that we should help workers, but we need to intervene earlier – in the hiring stage. Currently older workers cannot get a job, so we should prohibit employers from asking about age.

Advisor 3: I think we should set up a platform for job-seekers. We can find a private firm to construct the platform and then regulate the prices it charges.

Advisor 4: I have a different idea. Why don't we demand our banks to give loans to poor entrepreneurs, so they can just build their own business?

Discuss (briefly!) whether these four advices make sense from an economic perspective. **[25 points]**